



Igniters Innovation Challenge 2026 | Engineering Track

An exclusive Engineering Track

Hosted by: Suzuki R&D Center

TEAM WOOKIES

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Recommendation: validate before selling. A 30-car hydrogen retrofit pilot in a captive Faridabad fleet, scaled only if delivered hydrogen falls below about ₹190 per kg

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Answer to the brief: a CNG-to-hydrogen retrofit can be made safe by 2030, but at unsubsidised hydrogen prices it is not affordable; its 2030 role is a policy-supported niche of about 1,400 cars, not a mass product.

1. THE ANSWER TO THE BRIEF'S QUESTION

- **Safe:** yes, with known engineering (composite cylinder, thermally activated relief, twin leak sensors, lean burn with NOx control). But no approval route exists today for a hydrogen retrofit of a car; creating one is gate 1.
- **Affordable:** not by 2030. At ₹800/kg delivered (2028) the converted car costs ₹13.7/km all-in versus ₹5.2/km on CNG. Fuel parity with CNG needs ₹188/kg; the cheapest 2030 forecast found is about ₹360/kg delivered.
- **Viable complement:** only as a captive-fleet niche next to hydrogen hubs. A compact BEV bought in 2028 comes to ₹5.6/km including purchase, so BEVs stay the economic default for high-utilisation fleets.
- **So the strategy is an option, not a launch:** an HCNG step that needs no vehicle change, a bounded H2-ICE retrofit pilot that creates the approval route and the data, and a scale decision in 2030 against explicit price gates.

2. FIRST CUSTOMER AND USE CASE

- **Donor:** Maruti Suzuki Dzire Tour S CNG, 3 to 5 years old, factory bi-fuel; petrol stays as the automatic backup fuel.
- **Fleet:** captive, return-to-base institutional and contract fleets (PSU and corporate staff transport, airport contracts) in Faridabad, Delhi-NCR, running 150 to 210 km per day with two depot fills.
- **Why here:** IOCL R&D Faridabad hosts India's only car-capable hydrogen dispensing point, and the MNRE Sahibabad-Faridabad-Delhi hydrogen corridor (sanctioned March 2025) brings buses and trucks that a shared station needs.
- **Why not Delhi app-cabs first:** the 2023 aggregator scheme counts only electric cars toward onboarding ratios, and CAQM allows only CNG or EV new aggregator vehicles in NCR from January 2026. A hydrogen conversion earns no credit there.
- **Second hub:** Ahmedabad-Vadodara-Surat corridor (Gujarat funds 30 percent of the first 20 hydrogen stations).

3. HOW BIG, HONESTLY (bottom-up, 2030)

Step	Vehicles
CNG vehicles on road, Mar 2025 (all types)	8,195,000
Passenger cars among them (estimate)	4,507,250
Factory-fitted, boot cylinder, under 6 yr in 2028	2,253,625
In commercial or institutional fleets	450,725
In cities with a hydrogen hub by 2030	180,290
Plausible early adopters (hypothesis)	9,014
Station-limited capacity: 8 hubs x 174 cars	1,394
Realistic converted fleet by 2030	1,394

Round 1 said 1,00,000 cars by 2030. Stations, not demand, set the ceiling: one 500 kg/day hub at 70 percent utilisation serves about 174 cars.

Customer demand is a hypothesis: no fleet interviews exist yet; Phase 0 buys 30 of them.

Same idea as Round 1: convert the installed CNG base, fleets first, aggregate hydrogen demand at hubs. What changed: pump price instead of production cost, petrol backup instead of a second cylinder, a station-limited rollout, and explicit stop rules.

Evidence labels used throughout: verified = external source in the register; model = workbook output; assumption = stated input range; target = proposed value requiring validation.

H2-ICE is the only conversion at conversion cost, but it carries the weakest fuel economics; a fuel-cell conversion costs as much as a new BEV

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Matched criteria for the same donor and duty (Dzire Tour S CNG, 55,000 km per year, owner decision in 2028). Values are model outputs unless a source is named.

Criterion	H2-ICE retrofit (selected)	Fuel-cell conversion	New compact BEV (donor sold)	Continue on CNG
What changes	Cylinder, lines, injectors, ECU, sensors; engine and petrol system retained	Whole powertrain: stack, battery, motor, inverter, cooling, tank, controls	New vehicle; donor sold for about ₹3.5 lakh	Nothing
Cost to the owner	₹3.2 lakh installed at 10,000/yr; ₹7.7 lakh in the pilot	₹15 to 25 lakh pilot; ₹8 to 12 lakh at volume (estimate)	₹10.5 lakh purchase (Tiago EV to Punch EV class)	0
Energy cost per km, 2028	₹12.0 blended (hydrogen km ₹14.8 at ₹800/kg)	About half of H2-ICE per hydrogen km	₹1.5 (depot and public mix)	₹3.6
Range per fill or charge	61 km on hydrogen plus 555 km on petrol	About 2x H2-ICE on the same tank	187 to 260 km in real-world tests	About 250 km at 25 km/kg
Refuel or recharge time	3 to 5 min (350 bar, non-cooled fill)	3 to 5 min	About 1 h DC (10 to 80 percent); 6 to 8 h AC	3 to 5 min plus queue
Peak power versus petrol	About 52 percent (lean burn); city duty	Restored by electric drive	Higher	About 85 percent
Tailpipe	Zero CO2 on hydrogen km; NOx must be controlled	Water vapour only	Zero	About 105 g CO2/km
Approval route today	None for retrofits; AIS-195 covers new H2-ICE vehicles	None; AIS-157 scope excludes retrofitted or converted FCEVs	Established	Established
Supplier readiness	Injectors, valves, regulators production-intent; car-size cylinders scarce	Imported stacks; no light-duty supply chain in India	Mature	Mature

WHAT GLOBAL EXPERIENCE TEACHES

- **Captive fleets with depot fuelling survive:** KEYOU truck conversions (Germany, 2024 to 2027 with Daimler), ULEMCo van conversions (UK), Toyota's hydrogen HiAce at its Altona depot (2024), NTPC Leh buses (80 kg/day, 350 bar).
- **Retail light-duty networks retreat:** Shell closed its 7 California car stations (Feb 2024); H2 Mobility closed 36 German car stations (2025); Stellantis cancelled its hydrogen vans (Jul 2025); Toyota sold 210 Mirai in the US in 2025.
- **H2-ICE momentum is heavy-duty:** JCB EU type approval (May 2025); Cummins B6.7H built in Jamshedpur for Tata trucks (Mar 2025); 70 MNRE pilot vehicles are buses and trucks, zero cars.
- **Small-car hydrogen combustion is live but pre-commercial:** Suzuki and AVL hydrogen Swift prototype (1.4 L direct injection, 100 kW, Vienna, May 2026); Suzuki hydrogen Burgman (2023, 2025). No company sells a passenger-car hydrogen conversion anywhere.
- **Transfer to India:** design for a depot, not a network; treat fuel price as the gating variable; borrow heavy-duty supplier parts; do not cite prototypes as proof of cost.

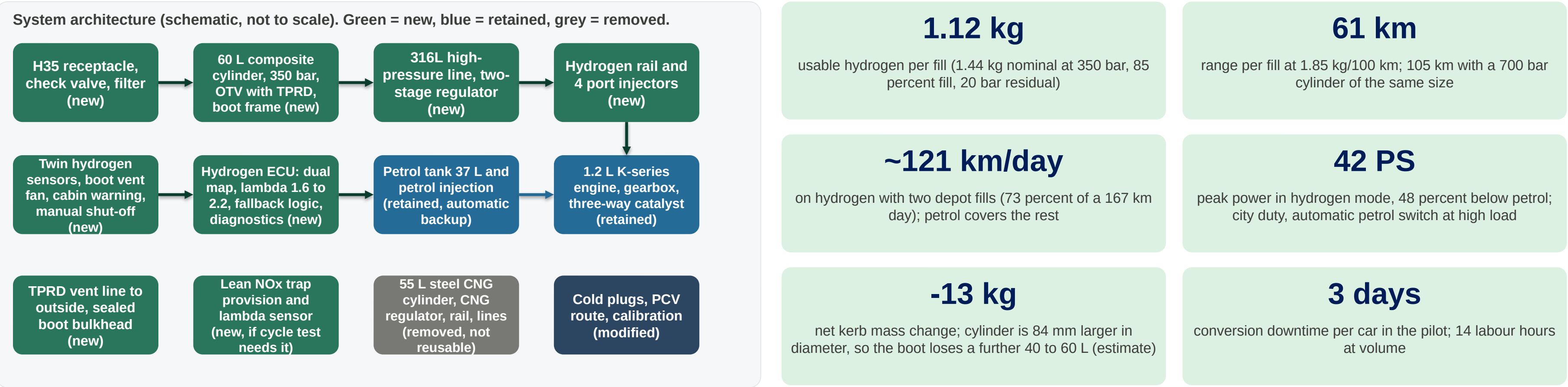
Selected route: H2-ICE with port injection, lean burn and automatic petrol fallback. It is the only architecture that is a retrofit in cost and scope; its case rests entirely on the delivered hydrogen price, so the pilot is designed to measure exactly that.

What the retrofit changes: storage, fuel delivery, controls and safety are new; the engine, petrol system and body are retained (concept, to be verified by mock-up)

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Donor: Maruti Suzuki Dzire Tour S CNG, 1.2 L K-series, 60 kW on petrol, 55 L steel CNG cylinder in the boot, 37 L petrol tank, 163 mm ground clearance (no under-floor cylinder option).



Safety: the hazards are known and controllable; approval is the real gap, because no route exists yet for a hydrogen retrofit of a car

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Hydrogen ignites between 4 and 75 percent in air at 0.017 mJ, diffuses 3.8 times faster than natural gas and burns almost invisibly: different risks, not automatically higher, if designed for.

Hazard	Prevention (design)	Detection and response	Validation evidence
Leak into boot, cabin, garage or basement	Sealed boot bulkhead, vented boot, no ignition sources in the bay, valve closed when engine off	Two hydrogen sensors (alarm at 1 and 2 percent), tank valve closes within 1 s, ECU switches to petrol, driver warning	Concentration below 4 percent in cabin and below 8 percent at vents (GTR 13 / AIS-195 type tests); enclosed-space test
Embrittlement, permeation, material ageing	Hydrogen-rated 316L lines, elastomers and composite liner; no reuse of CNG steel parts	3-yearly cylinder requalification (Gas Cylinder Rules 2025), 15-year life, leak check at service	Supplier qualification to ISO 19881 / UN R134; cycling and permeation tests
Crash damage	Cylinder behind rear seat inside the crush-free zone; frame designed to 20 g / 8 g loads	Inertia switch closes the tank valve; post-crash leak limit	Sled test of the mounting, full-vehicle rear crash under AIS-195 procedures
Overpressure and fire	TPRD vents away from occupants, wheel wells and electrics; burst ratio 2.25 x working pressure	TPRD opens on heat; controlled release rather than rupture	Bonfire, burst, drop and cycling tests (GTR 13); vent-line routing review
Detection or control failure	Redundant sensors, ECU watchdog, manual shut-off valve, limp-home on petrol	Fault codes, forced petrol mode, service lock-out	Hardware-in-the-loop and fault-injection tests; 500 h engine durability
Workshop and refuelling incidents	NFPA 2 type ventilation (1 cfm per sq ft), hydrogen detection, permit-to-work, PESO-trained staff	Gas detection interlocks on the dispenser and in the bay; nitrogen purge for venting	PESO Form-H licence for the dispensing point; SAE J2601 fill protocol; lessons from Kjørbo 2019 (assembly error) and Santa Clara 2019 (procedures)

NOx: hydrogen combustion still forms thermal NOx (US DOE). Lean burn above lambda 2 keeps engine-out NOx near zero at steady state, but accelerations leave that regime, so the design carries a lean NOx trap provision and must meet BS-VI 60 mg/km on the certification cycle. This is a target to demonstrate, not a result.

APPROVAL PATH AND OPEN QUESTIONS

- **Exists:** AIS-195:2023 (Amd 1, May 2025) for new M and N hydrogen ICE vehicles via CMVR Rule 125M (GSR 746(E), 16 Oct 2023).
- **Exists in principle:** MV Act s.52 and CMVR Rule 126 proviso (retrofitted vehicles tested and type-approved by test agencies) plus the Rule 115 CNG-kit machinery: kit type approval, authorised workshops, RC endorsement (Form 22D).
- **Gap:** no hydrogen retrofit kit standard or emission procedure; AIS-157 explicitly excludes retrofitted or converted fuel-cell vehicles; the AIS-195 position on retrofits could not be verified (document inaccessible in this study).
- **Cylinders and stations:** Gas Cylinder Rules 2025 now cover on-board hydrogen cylinders; PESO Form-H licences for dispensing online since May 2025; no car-size PESO-approved cylinder yet.
- **Trial route:** trade certificate plus test-agency evaluation, as used for the ICAT Mirai pilot (2022) and the Tata truck trials (2025); the MNRE transport pilot scheme names 4-wheelers.
- **Insurance and warranty:** IMT-25 covers only CNG and LPG kits; OEM warranty is void after conversion. Both need new instruments.

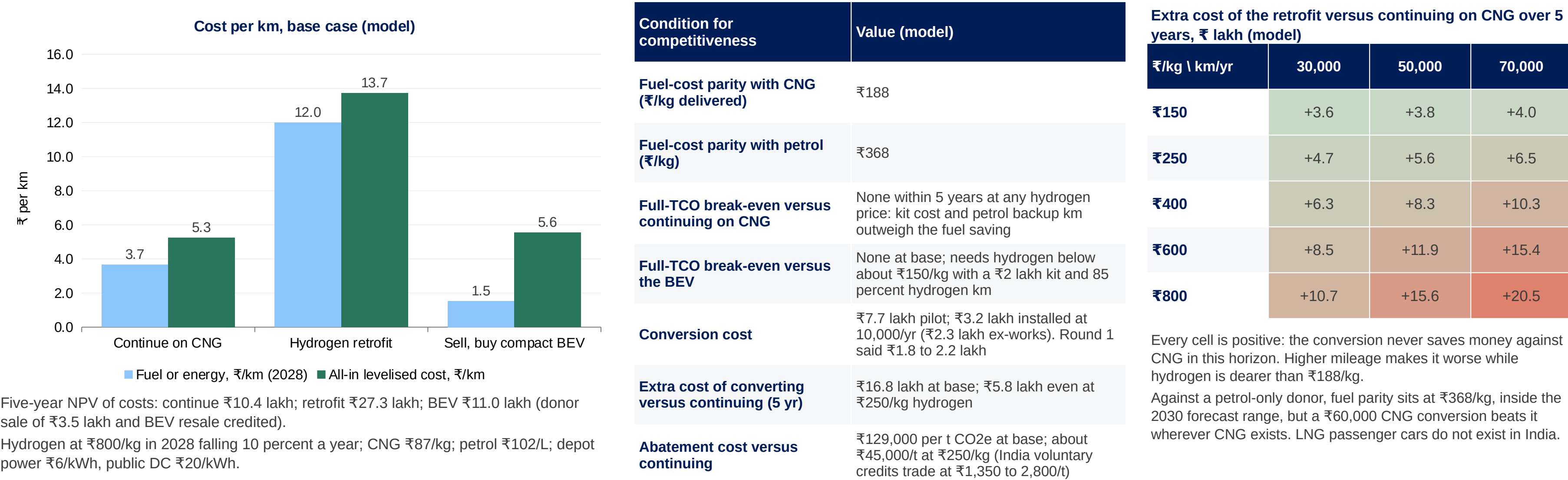
We do not claim 'approved' or 'fully safe'. Gate 1 (March 2027): a written regulator and test-agency view on the retrofit route, or the programme stops.

Economics: at every credible hydrogen price to 2032 the retrofit costs the owner more than staying on CNG; the sign never turns

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Fair comparison for an owner deciding in 2028 with a 3-year-old Dzire Tour S CNG: keep it, convert it, or sell it and buy a compact BEV. Five years, 55,000 km per year, 12 percent discount rate, unsubsidised, same time costs for queues and charging.



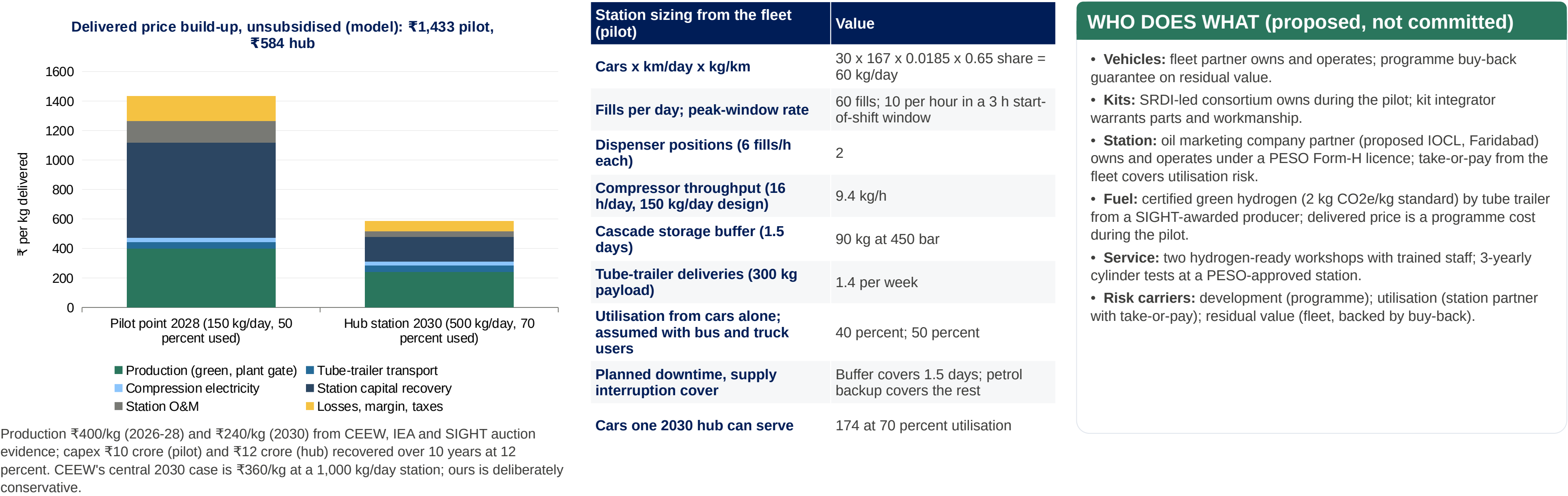
Reading: the retrofit is a decarbonisation cost, not a saving. It needs a paying counterparty (policy or a corporate abatement budget) or a far cheaper hydrogen molecule than any 2030 forecast found (CEEW ₹360/kg delivered; record refinery-gate tender ₹279/kg before compression and dispensing).

Fuel supply: a 30-car pilot needs about 60 kg a day; delivered hydrogen costs about ₹1,433/kg at a pilot point and about ₹584/kg at a 2030 hub at 70 percent utilisation

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Pilot setting: Faridabad, Delhi-NCR. IOCL R&D dispensed hydrogen for the 2022 Mirai pilot and its fuel-cell buses; the MNRE Sahibabad-Faridabad-Delhi corridor was sanctioned in March 2025 (18 to 24 months to commission). No station is public or confirmed available: access is a proposed partnership.



Utilisation alone cannot close the gap: at 95 percent utilisation the hub price is still about ₹524/kg, and CNG parity (₹188/kg) is out of reach because production, transport and compression alone exceed it. Delivered versus on-site electrolysis: delivered hydrogen keeps the pilot's capital to a dispensing point; an electrolyser is justified only at hub scale.

Roadmap to 2030: four gated phases, ₹37.6 crore including contingency, with a proceed, redesign or stop decision at each gate

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Calendar phases from October 2026; each phase ends with evidence, not a date. Suzuki R&D Center India is proposed as engineering lead; public support is sought under the MNRE transport pilot scheme, which names 4-wheelers.

PHASE 0 Oct 2026 to Mar 2027		PHASE 1 Apr 2027 to Mar 2028		PHASE 2 Apr 2028 to Mar 2029		PHASE 3 Apr 2029 to Sep 2030		PHASE 4 from Oct 2030	
Feasibility and validation ₹0.9 cr		Engineering and controlled validation ₹8.5 cr		Certification and pilot build ₹15.2 cr		Pilot operation and evaluation ₹8.1 cr		Expansion only if gate 4 passes conditional	
	Phase 0	Phase 1	Phase 2	Phase 3	Phase 4 (conditional)				
Deliverable	Written regulator and test-agency view on the retrofit route; 30 fleet interviews and telematics on 20 cars; boot mock-up and CAD packaging; station partner letter	500 h hydrogen engine test-cell programme; 3 converted prototypes; storage qualification (burst, cycling, bonfire, drop); vehicle leak, ventilation and crash-structure tests	Kit type approval under the agreed route; 80,000 km durability on 2 cars; 30 conversions; dispensing point commissioned; 2 workshops upgraded	18 months of operation with 10 CNG and 10 BEV comparison cars; quarterly emission tests; independent evaluation report	1,000 to 1,500 cars at up to 8 hub stations; localised cylinders and kit production				
Accountable	SRDI engineering lead; ARAI or ICAT; PESO liaison	SRDI with a test-cell partner (HORIBA Chakan or ARAI); cylinder supplier	Kit integrator; ARAI or ICAT; OMC station partner; fleet partner	Fleet partner operations; independent evaluator (IIT or CEEW); SRDI data team	Consortium and public programme				
Dependencies	Regulator engagement; fleet access	Car-size cylinder samples (Luxfer, Quantum, NPROXX, Time Technoplast); hydrogen injectors	MNRE pilot sanction; PESO approvals; corridor station timing	Fuel supply contract; insurer endorsement	Delivered hydrogen price; policy support				
Gate and decision	Gate 1: route defined, partner secured, demand evidence from 3 fleets. Otherwise stop.	Gate 2: consumption at or below 2.2 kg/100 km, NOx at or below 60 mg/km on cycle, no abnormal combustion in 500 h, safety tests passed. Otherwise redesign once, then stop.	Gate 3: approval granted and station commissioned within 6 months of plan. Otherwise pause.	Gate 4 (Sep 2030): availability 95 percent or better, zero reportable hydrogen incidents, measured ₹/km published, delivered hydrogen at or below ₹250/kg with a credible path to ₹190/kg. Otherwise stop and publish.	Proceed only with fuel price and approval in place				

Proposed funding split (not committed)	Share	₹ crore
Suzuki R&D Center India and Maruti Suzuki: engineering, prototypes, test cells	40 percent	15.0
Public support sought: MNRE transport pilot scheme (4-wheelers named) or R&D scheme, state hydrogen policy	35 percent	13.1
Oil marketing company partner: dispensing point	15 percent	5.6
Fleet partner: vehicles, drivers, operations	10 percent	3.8
Total to September 2030 (incl. 15 percent contingency)	100 percent	37.6

Why Suzuki's role is plausible: CNG engine and calibration know-how, a 70 percent share of India's CNG cars, and its own hydrogen combustion work (Swift prototype with AVL, 2026; Burgman). This is a proposal; no endorsement is implied.

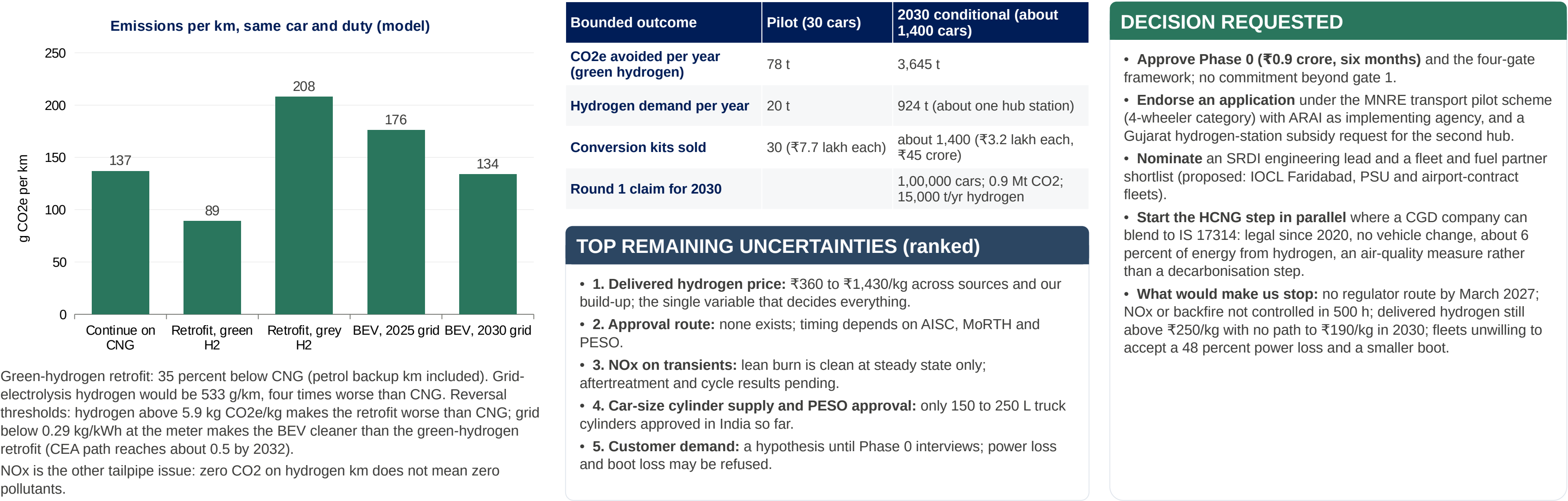
Sources: MNRE scheme guidelines, PIB 14 Feb 2024 (buses, trucks and 4-wheelers) [PIB PRID 2006052]; ARAI Phase-II call for proposals (Sep to Dec 2025); HORIBA and ARAI hydrogen test facilities [W4-S14, S15]; CNG kit approval precedent [W2-S3, S8]; PESO rules [W2-S24]; model workbook (Pilot_Budget).

Expected impact is real but small by 2030; the decision requested is ₹0.9 crore for Phase 0, which buys the facts that decide whether ₹37.6 crore is worth spending

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Bounded outcomes at matched service (same car, same duty), with the assumptions under which each conclusion reverses.



H2-SHIFT keeps India's hydrogen retrofit option open at low cost, tells the truth about price, and hands the 2030 scale decision to evidence instead of ambition.

Key sources and annexure index

Research cut-off 14 September 2026. Full register of 300 plus entries with URLs, dates, sections and limitations is in the annexure and the [workbook \(Sources sheet\)](#).

1. PIB, MNRE, 'Pilot Projects on Hydrogen Fuelled Buses and Trucks Launched under the NGHM', 3 Mar 2025 (PRID 2107795): 5 projects, 37 vehicles, 9 stations, 10 routes incl. Sahibabad-Faridabad-Delhi, ₹208 crore.

2. PIB, MNRE, 'Scheme Guidelines for Pilot Projects on use of Green Hydrogen in the Transport Sector', 14 Feb 2024 (PRID 2006052): buses, trucks and 4-wheelers; ₹496 crore; PIB 29 Jul 2026: 12 projects, 70 vehicles, 16 stations, ₹410.77 crore.

3. MoRTH, GSR 746(E), 16 Oct 2023: CMVR Rule 125M, AIS 195:2023 for hydrogen ICE vehicles of categories M and N; ARAI AIS-195 with Amendment 1 (May 2025); AIS 157:2020 (GSR 579(E), 23 Sep 2020), scope excludes retrofitted or converted fuel-cell vehicles.

4. DPIIT/PESO, Gas Cylinders (Amendment) Rules 2025, GSR 225(E), 11 Apr 2025: on-board hydrogen cylinders, 3-yearly testing, 15-year life; PESO Form-H online licensing for hydrogen dispensing, May 2025.

5. MoRTH, GSR 585(E), 25 Sep 2020: H-CNG per IS 17314:2019 added to CMVR Rule 115; IOCL and DTC 50-bus HCNG trial, Oct 2020.

6. PNGRB citing Vahan (Aug 2025): 81.95 lakh CNG vehicles (Mar 2025); MoPNG: 8,150 plus CNG stations, 17,700 to 18,000 targeted by 2030; IGL, MGL, MNGL price notices Sep 2026.

7. Maruti Suzuki specifications and releases (Tour S, Dzire, WagonR, Ertiga, Eeco); Maruti CNG sales FY25 6.12 to 6.2 lakh, FY26 above 7 lakh; Suzuki Growth Strategy FY2030.

8. Luxfer G-Stor Pro H2 specification table (V060H35: 60 L, 350 bar, 36 kg, 1.45 kg); Hexagon Purus Type 4 table (Rev 08/22); US DOE Records 19008 and 24006 on Type IV storage cost.

9. US DOE and H2Tools: hydrogen properties; 'Does the use of hydrogen produce NOx'; NFPA 2 (2023); UN GTR 13 Amendment 1 (2023) and UN R134; SAE J2601 (2020); ISO 19881:2025; ISO 14687.

10. CEEW (2025) India hydrogen and EV TCO model: delivered green hydrogen USD 4.2/kg in 2030; ICCT (2022, 2025): station cost by utilisation, lifecycle factors; IEA Global Hydrogen Review 2025; NITI Aayog (Feb 2026) taxi mileage 70,000 km/yr.

11. CEA CO2 Baseline Database v21 (FY2024-25, 0.710 t/MWh) and National Electricity Plan 2022-32; IPCC 2006 fuel factors; MNRE Green Hydrogen Standard (2 kg CO2e/kg).

12. Global programmes: Toyota (Mirai, HiAce, GR Corolla), Hyundai Nexa, KEYOU, ULEMCo, JCB, Cummins B6.7H India, Stellantis (Jul 2025), Shell California (Feb 2024), H2 Mobility Germany (2025); Suzuki-AVL hydrogen Swift, Vienna Motor Symposium, May 2026 (Fuel Cells Works, RushLane).

13. Delhi Motor Vehicle Aggregator and Delivery Service Provider Scheme 2023; Delhi EV Policy 2026; CAQM NCR direction (Jan 2026); Gujarat Green Hydrogen Policy 2025; Maharashtra Green Hydrogen Policy 2023.

ANNEXURE INDEX (separate PDF and workbook)

- A1 Claim audit: Round 1 versus evidence, with corrections

A2 Assumptions, evidence classes and full source register

A3 Cost model: BOM, TCO, break-even, sensitivities

A4 Packaging, energy balance and performance calculations

A5 Regulatory compliance matrix and hazard analysis

A6 Market sizing method and bottom-up funnel

A7 Global benchmark evidence (progress and setbacks)

A8 Pilot design, instrumentation and first 90 days

A9 Phase budgets, responsibilities and funding

A10 Judge Q&A, validation log and change log

Workbook: H2SHIFT_model.xlsx (11 sheets, formula-driven, recalculated)

Evidence discipline: every headline number traces to a source ID (verified), a workbook cell (model) or a stated assumption. Many original documents were inaccessible through the build environment's network policy; such entries are marked 'excerpt' or 'summary' in the register and should be re-verified before external use. No customer interviews, letters of interest, test results or partnerships are claimed.